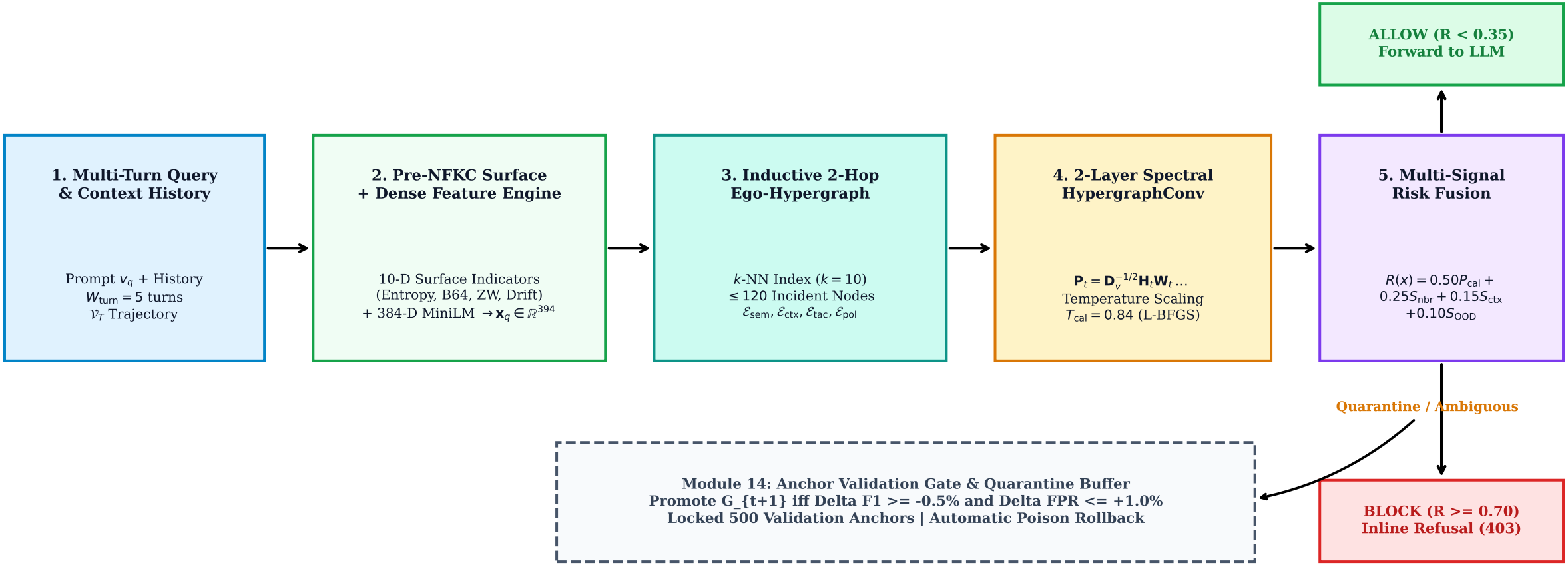

AHD-Jailbreak: Inductive Pre-Inference Defense Architecture & Anchor-Gated Adaptation Pipeline
1. Multi-Turn Query & Context History
Prompt v_q + History
$W_{\text{turn}} = 5$ turns
$\mathcal{V}_T$ Trajectory
2. Pre-NFKC Surface + Dense Feature Engine
10-D Surface Indicators
(Entropy, B64, ZW, Drift)
+ 384-D MiniLM $\rightarrow \mathbf{x}_q \in \mathbb{R}^{394}$
3. Inductive 2-Hop Ego-Hypergraph
k-NN Index ($k = 10$)
≤ 120 Incident Nodes
$\mathcal{E}_{\text{sem}}, \mathcal{E}_{\text{ctx}}, \mathcal{E}_{\text{tac}}, \mathcal{E}_{\text{pol}}$
4. 2-Layer Spectral HypergraphConv
$\mathbf{P}_t = \mathbf{D}_v^{-1/2}\mathbf{H}_t\mathbf{W}_t \ldots$
Temperature Scaling
$T_{\text{cal}} = 0.84$ (L-BFGS)
5. Multi-Signal Risk Fusion
$R(x) = 0.50P_{\text{cal}} +$
$0.25S_{\text{nbr}} + 0.15S_{\text{ctx}}$
$+0.10S_{\text{OOD}}$
ALLOW (R < 0.35)
Forward to LLM
Quarantine / Ambiguous
BLOCK (R >= 0.70)
Inline Refusal (403)
Module 14: Anchor Validation Gate & Quarantine Buffer
Promote G_{t+1} iff Delta F1 >= -0.5% and Delta FPR <= +1.0%
Locked 500 Validation Anchors | Automatic Poison Rollback